

Bohan Lu

Education

- 2023–2027 **PhD Candidate in Electrical and Computer Engineering**, *Duke University, USA*.
- Winter 2019–Summer 2022 **MSc in Physics**, *Freie Universität Berlin, Germany*.
- 2022
 - Master thesis: Entanglement dynamics of a many-body localized system
- Fall 2015–Spring 2019 **BA in Mathematics and Physics**, *Washington University in St. Louis, USA*.
- Dean's List Fall 2015–Fall 2016 and Spring 2019
 - Cum Laude
 - Distinction in Mathematics
 - High Distinction in Physics
 - Bachelor thesis in physics: Reduced density operators of finite-sized Heisenberg XXZ chains

Research Experience

- July–October 2022 **Research Internship**, *HQS Quantum Simulations GmbH, Karlsruhe, Germany*.
- Conducted three-month research on developing a novel spectral resolution method based on Hermite polynomials for resonant-level models, supervised by Dr. Peter Schmitteckert
 - Programmed and benchmarked such method and composed an internal report
- April 2021–June 2022 **Master Thesis**, *Physics Department, Freie Universität Berlin, Germany*.
- Conducted research on extracting information on the local integrals of motion of a many-body localized system from its entanglement dynamics, supervised by Professor Jens Eisert
 - Proposed a protocol to extract measurable quantities from a theoretical toy model
 - Performed tensor network simulations on clusters and analyzed quantum information bounds for entanglement growth
 - Thesis available on my website and resulted in a preprint: [arXiv 2301.01787](https://arxiv.org/abs/2301.01787) (first author with three co-authors)
- Fall 2017–Summer 2018 **Greg Delos Fellowship**, *Physics Department at Washington University*.
- Conducted research under Professor Alexander Seidel on the space of reduced density operators of ground states of the finite-sized XXZ Heisenberg chain, resulting in a presentation at APS March Meeting 2019
 - Computed the convex set of ground states of a family of the Heisenberg XXZ Hamiltonian with Bethe Ansatz and DMRG methods
 - Searched for signatures of phase transition by associating the convex set manifold of finite-sized systems with a support vector machine
- Fall 2016–Summer 2017 **Advanced Research Training for Undergraduates**, *Math Department at Washington University*.
- Performed independent study on topology and eight-week research under Professor Tang Xiang on the generalization of Gauss-Bonnet theorem to two-dimensional closed orbifolds;
 - Presented at the Undergraduate Research Symposium.
- Summer 2016 **Summer Fellowships in Physics**, *Physics Department at Washington University*.
- Conducted ten-week research under Dr. Manel Errando and Professor Henric Krawczynski on the search for extended halos around TeV blazars using maximum-likelihood inference methods.
 - Presented at the Undergraduate Research Symposium.

Publications

- 2023 **Nature Communications Physics**, *Measuring out quasi-local integrals of motion from entanglement*, Authors: Bohan Lu, Christian Bertoni, Steven Thomson and Jens Eisert, <https://arxiv.org/abs/2301.01787>.

Talks

- 2019 **APS March Meeting**, *Reduced density operators of finite-sized Heisenberg XXZ chains*, Authors: Bohan Lu, Alexander Seidel.

Poster Presentations

- 2022 **CRC 183 Summer School on Many-body physics with Rydberg atoms**, *Entanglement dynamics in a many-body localized system*, Authors: Bohan Lu, Christian Berton, Steven Thomson and Jens Eisert.
- 2023 **Conference on Quantum Information Processing (QIP)**, *Improved decoding of quantum LDPC codes using neural-network enhanced belief propagation*, Authors: Bohan Lu, Arthur Pesah, Joschka Roffe and Nithin Raveendran.
- 2023 **QEC 2023 Sydney**, *Improved decoding of quantum LDPC codes using neural-network enhanced belief propagation*, Authors: Bohan Lu, Arthur Pesah, Joschka Roffe and Nithin Raveendran.
- 2025 **Conference on Quantum Information Processing (QIP)**, *A dynamical error-correcting code from a minimally non-local hypergraph product code*, Authors: Bohan Lu, Kenneth Brown.

Teaching

- Fall 2024 **ECE 520 Quantum Engineering TA**, *Pratt School of Engineering at Duke University*.
- Winter 2020 **Tensor network seminar TA**, *Physics Department at Freie Universität Berlin*.
- Fall 2017-Spring 2018 **Physics Help Desk Mentor**, *The Learning Center at Washington University*.
- Spring 2017 **Physics 216 Introduction to Special Relativity TA**, *Physics Department at Washington University*.
- Fall 2016 **Society of Physics Students Treasurer**, *Society of Physics Students at Washington University*.
- Fall 2016 **Math 4111 Introduction to Analysis Grader**, *Math Department at Washington University*.

Technical Skills

High performance computing, *Proficient with running jobs on clusters.*

Python, *Proficient with numerical libraries and basic software engineering.*

C++, *Basic competence.*

PyTorch, *Basic competence.*